Small-x factor base on NIST P-192: exact two-point relation yield

Study `research/prime_small_x_p192_20261014` · run 2026-10-14 · seed 7 · `ic prime` · `src/cryptanalysis/prime_small_x.rs`

1. Question and answer

Question. What is the probability, per trial, that a random point of NIST P-192 is a signed sum of two points of a small-x factor base, and how does the measurement compare with the estimates given before it was run?

Answer. With a base of B abscissae (2B points) the probability is **exactly $2B^2/(N-1)$** on P-192, where $N \approx 2^{192}$ is the group order: $2^{-175.0}$ at B = 256, $2^{-171.0}$ at 1024, $2^{-169.0}$ at 2048, $2^{-167.0}$ at 4096 and $2^{-165.0}$ at 8192, the largest base the tool builds. This is a count, not an estimate: every one of the up to 67 million pair sums was computed and none coincide, so no collision removes any. The expected number of trials per relation is the reciprocal, 2^{165} or more, and 2^{178} trials to collect B + 1 relations at B = 8192. Pollard rho costs about 2^{96} group operations. Monte Carlo on P-192 finds nothing, as it must, because the yield is 2^{146} times smaller than the 2M-trial run can resolve. The method is checked on toy primes, where the count equals the Monte Carlo rate to within 0.5% and the detector fires on every planted decomposition.

Against the estimates. The earlier estimate for a 2^{13}-point base was a yield near 2^{-166} and about 2^{-144} expected collisions. The measured yield for 8192 points (B = 4096) is $2^{-167.0}$, a factor of two lower than estimated. The estimate counted unordered pairs of points (2^{25}), but a sum and its negative have the same abscissa, so those pairs give 2^{24} distinct abscissae. The collision estimate stands: about 2^{-145} expected, 0 observed, which cannot discriminate. The estimates for a hypothetical 2^{40}-point base and for three-point sums were not measured.

2. Method

A small-x base takes the points with the smallest abscissae x = 0, 1, 2, … that lie on $y^2 = x^3 - 3x + b$, one sign of each, and uses both signs. Let the base have B abscissae. A signed sum of two base points has abscissa x(a + b) or x(a − b) for a pair of distinct base abscissae, or x(2a). There are B(B − 1)/2 pairs with two sums each and B doublings, which is B^2 abscissae if no two coincide. A target T is a signed sum of two base points exactly when x(T) is one of them, and then so is −T. The number of covered targets is therefore twice the number D of distinct sum abscissae, and a uniformly random group element decomposes in one trial with probability exactly

`yield = 2·D / (N − 1)` and $D \le B^2$, with equality when no two sums collide

`ic prime` computes D from every pair sum. The abscissa of a ± b is $\lambda^2 - x_1 - x_2$ with $\lambda = (y_1 \mp y_2)/(x_1 - x_2)$, and one inversion is shared by a whole row (Montgomery's trick), about seven field multiplications per pair. The keys are exact 192-bit field elements, sorted and deduplicated, so D is exact. P-192 uses the repository's native three-limb Montgomery field; toy curves use a 64-bit field. Because the count is exact, no target is sampled.

Three checks run with every measurement. **Monte Carlo**: random group elements (a random abscissa with a square right-hand side) are looked up in the key set, and the first hits are recovered as real decompositions by subtracting base points. **Planted controls**: a random signed sum of two base points is built by the group law and must be detected by the same lookup and recovered. **Toy curves**, where hits are common: $y^2 = x^3 - 3x + b$ over a prime $p \equiv 3 \pmod 4$ just below 2^{bits}, with b chosen so the cubic has no root (no rational 2-torsion, so every abscissa carries two points) and the order counted exactly by walking the field.

3. Results

3.1. NIST P-192

B	points	pair sums	collisions	exact yield	trials per relation	B + 1 relations	pair sums, s	MC µs/trial
256	512	2^{16}	0	$2^{-175.00}$	$2^{175.00}$	$2^{183.01}$	0.01	9.18
1024	2048	2^{20}	0	$2^{-171.00}$	$2^{171.00}$	$2^{181.00}$	0.10	9.17
2048	4096	2^{22}	0	$2^{-169.00}$	$2^{169.00}$	$2^{180.00}$	0.39	9.44
4096	8192	2^{24}	0	$2^{-167.00}$	$2^{167.00}$	$2^{179.00}$	1.70	10.11
8192	16384	2^{26}	0	$2^{-165.00}$	$2^{165.00}$	$2^{178.00}$	6.01	9.59

Table 1: P-192 small-x bases (largest abscissa x = 508 to 16535). Exact yield is 2D/(N − 1), identical to the collision-free model $2B^2/(N-1)$ in every row. "B + 1 relations" is $\log_2((B+1)/\text{yield})$. Monte Carlo: 2 000 000 trials, 0 hits in each run; wall microseconds per trial on 4 threads, about 4× that on one thread. Planted controls: 32 of 32 detected and 32 of 32 recovered in every run. Source: `data/p192_*.json`.

On P-192 the Monte Carlo is a null result: 0 hits in 2 000 000 trials bounds the rate above by 1.84×10^{-6} (exact one-sided 95% bound), and the exact yield is 2^{-165} at best. It cannot test the count; the toy runs do. What it does show is the cost of a trial at this size, about 38 µs on one thread for a random-point draw plus a lookup, which excludes the cost of deciding a target by solving the Semaev S_3 system.

3.2. Method check on toy primes

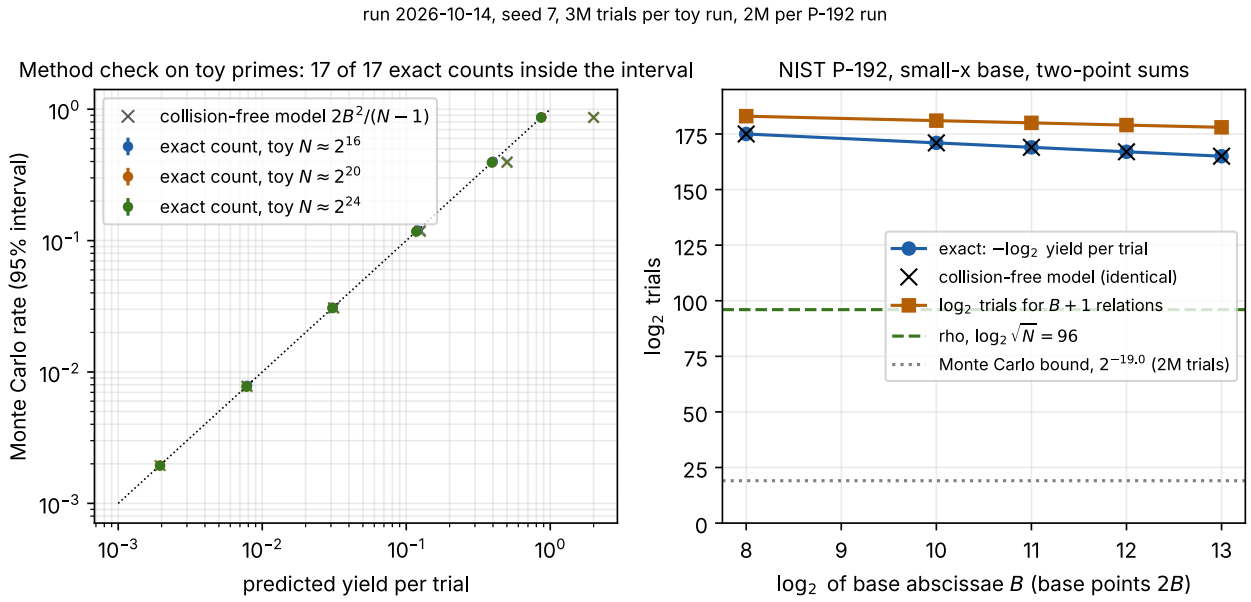


Figure 1: Left: Monte Carlo rate against the exact count (dots, with 95% intervals) and against the collision-free model (crosses), for 17 toy runs with 3 000 000 trials each. Right: P-192, base size against $\log_2$ trials. Run 2026-10-14, seed 7. Source: `data/*.json`; plot `graphs/plot_prime.py`.

In all 17 toy runs the exact count lies inside the 95% interval of the Monte Carlo rate, and the ratio of the two is between 0.993 and 1.002 (the spread is what the number of hits predicts: 5800 hits resolves 1.3%). The collision-free model is not close where collisions are common: at yield 0.87 (B = 4096, N ≈ 2^{24}) the model gives 2.0, and at yield 0.12 it is 5 to 6% high. That is the regime the exact count exists for. On P-192 the two coincide to every digit, because with B ≤ 8192 the expected number of collisions among the sums is below 2^{-140}. In every toy run, 32 of 32 planted decompositions were detected and recovered, and the first 8 Monte Carlo hits were recovered as real decompositions.

N	B	collisions	exact	model	MC rate	hits	MC / exact
65173	16	0	0.007856	0.007856	0.007852	23 557	0.9995
65173	32	17	0.03090	0.03142	0.03085	92 563	0.9984
65173	64	198	0.1196	0.1257	0.1197	358 976	1.0003
65173	128	3384	0.3989	0.5028	0.3988	1 196 271	0.9995
65173	256	37 316	0.8660	2.011	0.8659	2 597 822	0.9999
1 047 267	32	3	0.001950	0.001956	0.001940	5821	0.9951
1 047 267	128	298	0.03072	0.03129	0.03077	92 301	1.0015
1 047 267	512	55 745	0.3942	0.5006	0.3943	1 182 896	1.0003
1 047 267	1024	594 877	0.8664	2.003	0.8663	2 598 783	0.9998
16 777 897	128	15	0.001951	0.001953	0.001938	5814	0.9932
16 777 897	1024	63 063	0.1175	0.1250	0.1172	351 602	0.9976
16 777 897	2048	895 288	0.3933	0.5000	0.3928	1 178 290	0.9987
16 777 897	4096	9 523 333	0.8647	2.000	0.8649	2 594 613	1.0002

Table 2: Thirteen of the 17 toy runs; all 17 are in `data/`. Yields are probabilities per trial; the model is $2B^2/(N-1)$. Source: `data/toy*.json`.

4. Checks

- **Brute force.** On a 14-bit toy curve, every sum of two signed base points by the full group law reaches exactly twice as many distinct points as there are distinct sum abscissae (unit test).
- **Independent arithmetic.** On P-192, every batched sum abscissa of a ± b for the first six base points is found among the abscissae of the native Jacobian addition of the same points (unit test).
- **Monte Carlo.** On toy primes, 17 of 17 exact counts lie inside the 95% interval.
- **Controls.** Planted decompositions were detected and recovered in all 22 runs.
- **Counts.** The toy group orders equal the number of points found by walking the field, plus the point at infinity (unit test). The P-192 order is the standard prime N.

- **Inspection.** `ic inspect --curve secp192r1` (also `ic secp192r1`) passes the screens for the order, the modulus, nonsingularity and the generator, and the generator's order. Its cardinality check is reported as not implemented for prime curves.

5. Limits

1. **Two-point sums only.** A relation here is a signed sum of two small-x points. Three-point sums, which are what a prime-field attack would use, are not measured, and neither is the 2^{40}-point base of the earlier estimate.
2. **Yield, not solving.** The per-trial cost of deciding a target (a Semaev S_3 root find over the base) is not part of this measurement, and no linear algebra or descent runs. A relation yield is not a discrete logarithm, and the 2^{178} trials in the last column are a count of trials, not a time.
3. **One base family.** Only the small-x base exists. The tool says so. Other families (Kummer-tower, Dickson, structured bases) are not offered for prime fields.
4. **Size.** The tool builds at most 8192 abscissae because the pair sums are held in memory (67 million 192-bit keys at the maximum). At B = 8192 the yield is 2^{-165}, and the base would need about 2^{96} abscissae for $2B^2/N$ to reach one.
5. **Toy curves are not prime order.** They have odd, not necessarily prime, order and no rational 2-torsion. They validate the counting method and the Monte Carlo, not any property of P-192.
6. **The greedy gain.** The coverage-driven selector of the binary-field study chooses among Frobenius orbits; P-192 has none, and with zero collisions at every base size here a coverage-driven choice has nothing to improve: any base of the same size has the same yield, $2B^2/(N-1)$, provided its sums are distinct.

6. Reproduction

```
cargo run --release --bin ic -- --json prime --size 4096 --trials 2000000 --controls 32 --seed 7
cargo run --release --bin ic -- --json prime --curve toy:20 --size 512 --trials 3000000 --controls
32 --seed 7
cargo run --release --bin ic -- inspect --curve secp192r1
python3 research/prime_small_x_p192_20261014/graphs/plot_prime.py
```

`data/COMMANDS.txt` lists every run. Files: `data/*.json` (one `ic prime` report per run, 22 runs), `graphs/plot_prime.py`, `graphs/fig1_yield.{svg,png}`.